

NextStep Hacks 2026

EARTH FORWARD

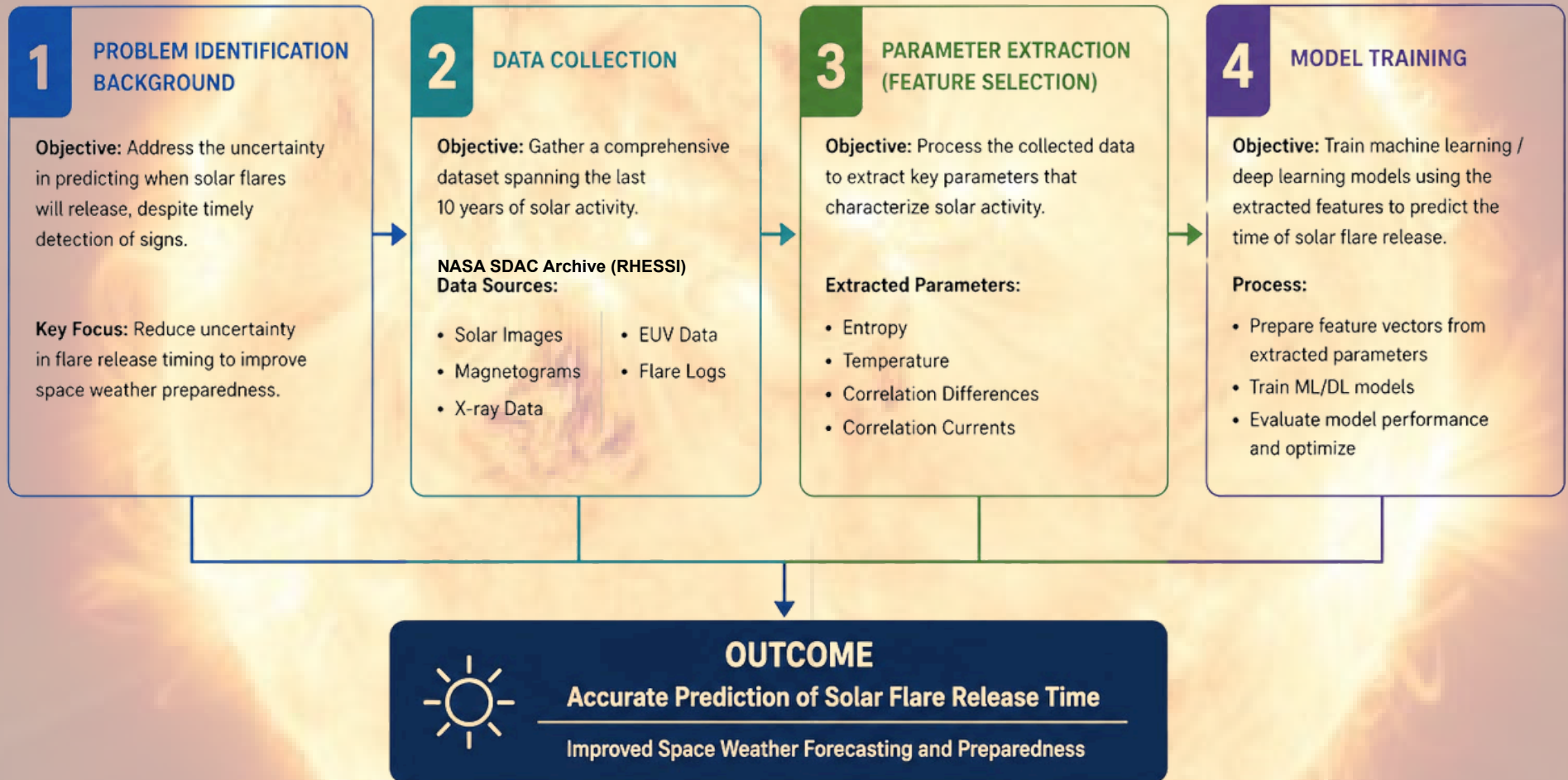


SunFLAAR is an **open-source Python package** for comprehensive solar flare analysis, integrating deep learning and statistical methods.

Team Name	High On Caffeine
Problem Theme	Earth Forward
Institute Name	Thapar Institute of Engineering & Technology
Team Members	Jashanpreet Singh Dingra, Tegkirat Kaur
Contact	+91 8847242013 (Team Leader)

Date: 16th September 2026

SOLAR FLARE PREDICTION PIPELINE



Problem Statement: Forecasting of Solar Flares using combined Soft and Hard X-ray data from Different Space and Ground Observatories

Solution: An open-source Python package called SunFLAAR for comprehensive solar flare analysis, integrating deep learning and statistical methods. It provides automated tools for multivariate time-series forecasting, active region classification, and imbalanced data metrics to aid space weather research.



3. Objectives

Web Portal

CLIMATOGRAPHY RESEARCHERS

An interactive, visual interface to map and analyze correlations between solar activity and long-term climatological data without requiring coding expertise.

Access Portal →

<https://sunflaar.streamlit.app>

REST API

SYSTEMS ENGINEERS

Integrate solar flare prediction capabilities and data streams effortlessly into custom applications or dashboards via documented endpoints.

Read Documentation →

<https://sunflaar.vercel.app>

Python Package

SOLAR PHYSICISTS

Direct programmatic integration of deep learning models, robust evaluation metrics, and MVTs preprocessing pipelines into existing Python workflows.

View on PyPI →

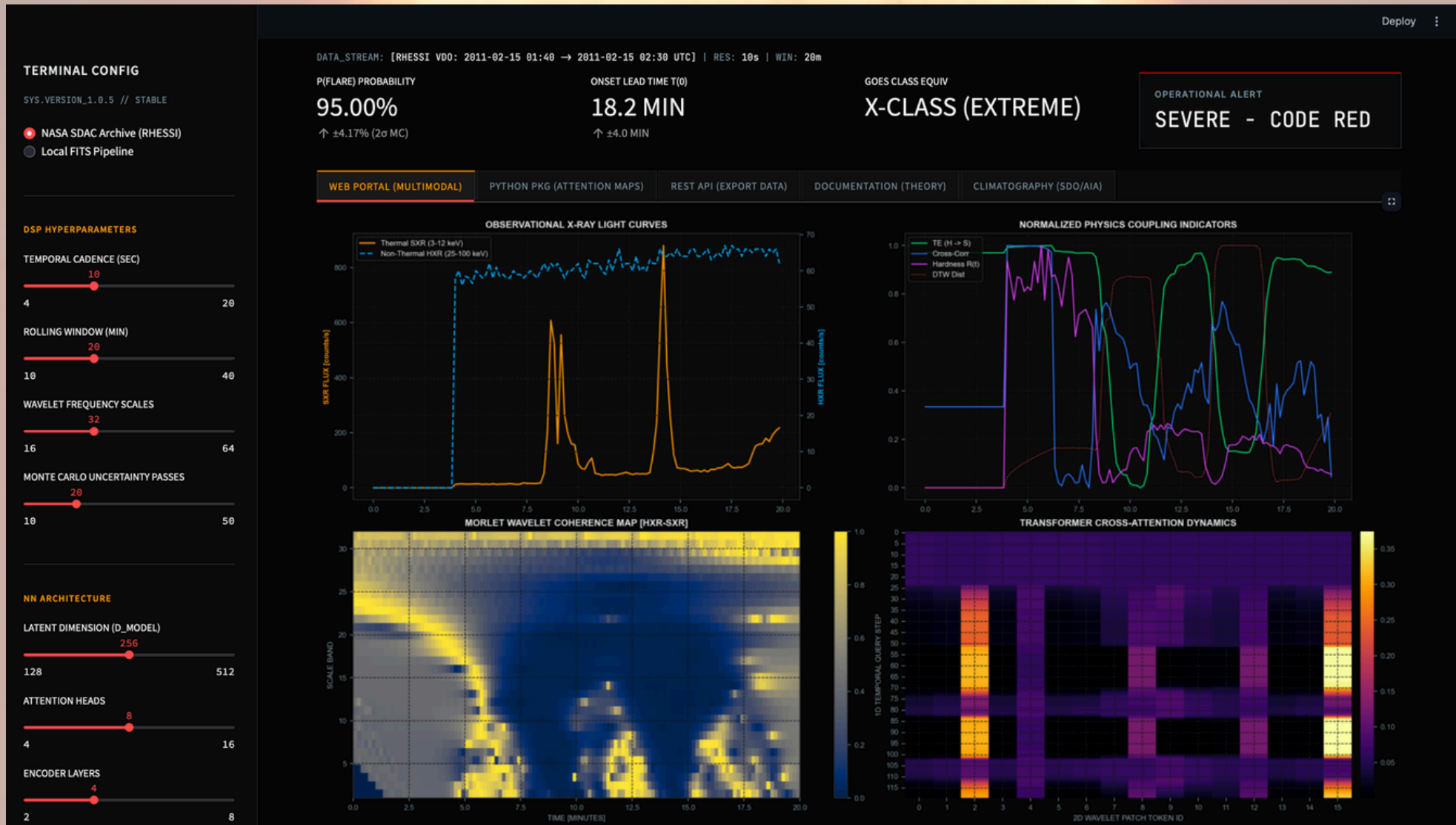
<https://pypi.org/project/sunflaar/>

To develop an open-source Python package that provides an integrated framework for automated solar flare analysis and forecasting, combining deep learning, multivariate time-series modeling, active-region classification, and robust statistical evaluation to support reproducible and accessible space weather research.



4. Proposed Solution

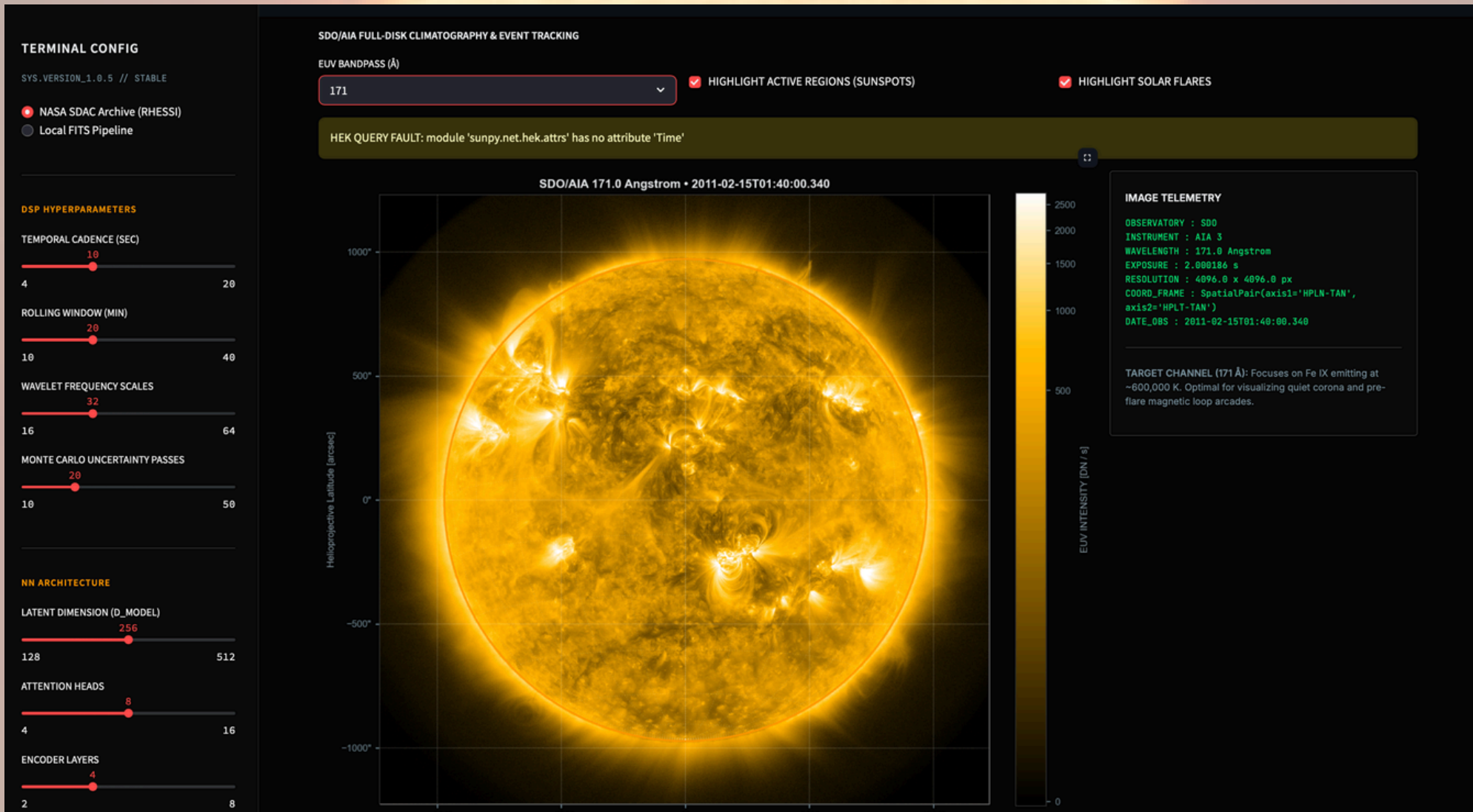
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Web Portal - Climate Scientists - <https://sunflaar.streamlit.app>

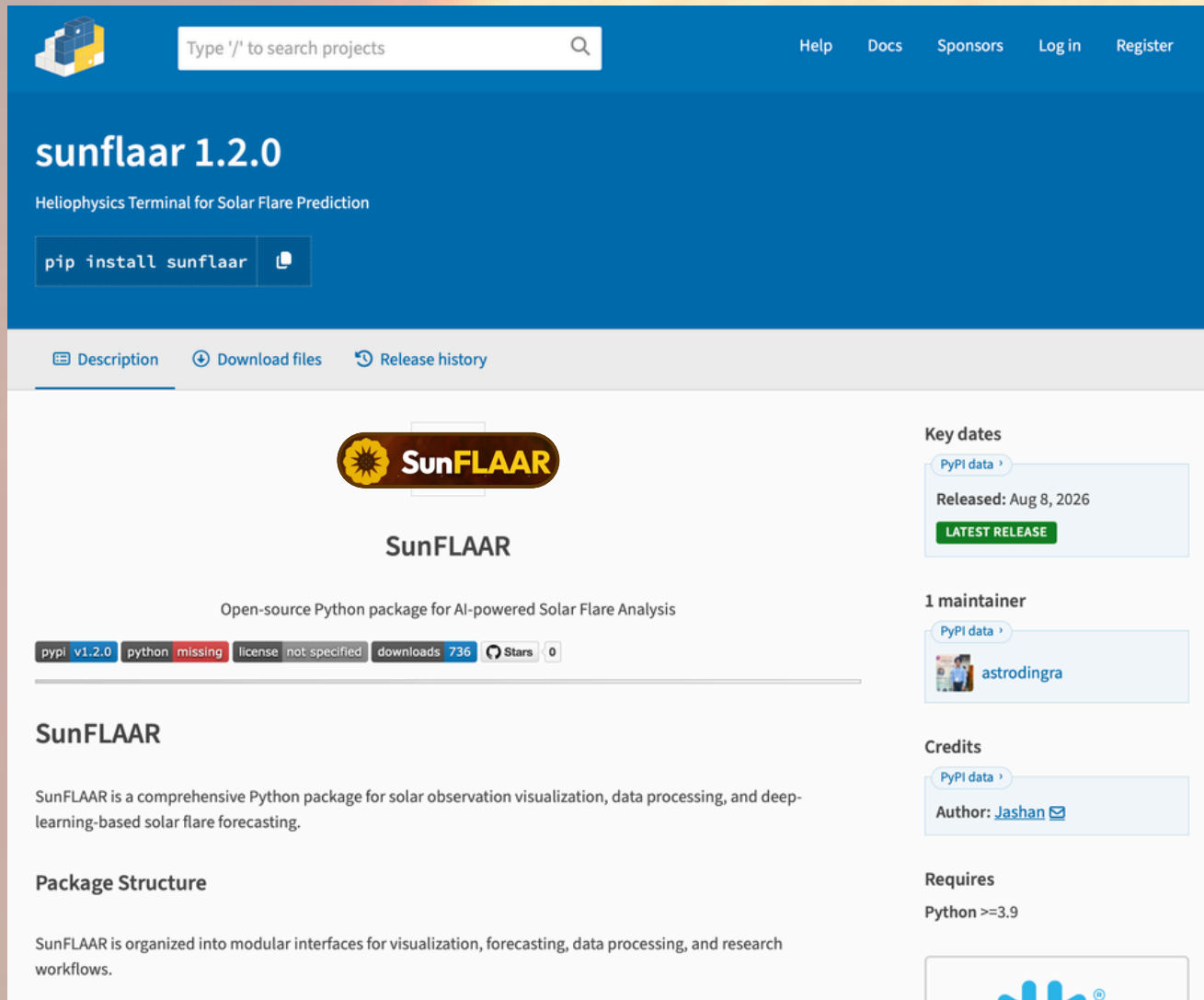
4. Proposed Solution

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Web Portal – Climate Scientists - <https://sunflaar.streamlit.app>

4. Proposed Solution



The screenshot shows the PyPI project page for **sunflaar 1.2.0**, described as a "Heliophysics Terminal for Solar Flare Prediction". The page features a blue header with a search bar and navigation links (Help, Docs, Sponsors, Log in, Register). Below the header, the package name and version are prominently displayed, along with a "pip install sunflaar" button. The main content area includes tabs for Description, Download files, and Release history. The description section features the SunFLAAR logo and states it is an "Open-source Python package for AI-powered Solar Flare Analysis". It also shows metadata such as "python missing", "license not specified", "downloads 736", and "Stars 0". The package structure is outlined as being organized into modular interfaces for visualization, forecasting, data processing, and research workflows. On the right side, key dates (Released: Aug 8, 2026), a single maintainer (astrodingra), and credits (Author: Jashan) are listed.

Sunflaar is a flexible Python package designed for solar physicists seeking a comprehensive tool for solar flare analysis. By simply running ***pip install sunflaar***, researchers can download the package to perform detailed data analysis, predict solar flares using our integrated deep learning models, and gain flexible, programmatic access to all core functions to support their research workflows.

PYPI Package – Solar Physicists

4. Proposed Solution

RESPONSE JSON

```
{
  "status": "operational",
  "version": "1.0.4",
  "uptime_seconds": 342190,
  "models_loaded": ["lstm_mvts_v2", "cnn_2d_baseline"]
}
```

POST /api/v1/predict

Submits Multivariate Time Series (MVTs) data from SHARP parameters for a specific Active Region (AR) to receive a flare classification probability.

Request Parameters

PARAMETER	TYPE	DESCRIPTION
ar_number	integer	NOAA Active Region identifier.
model_type	string	The predictive model to utilize (e.g., "lstm_mvts_v2").
features	array[string]	List of SHARP parameters corresponding to columns in time_series_data.
time_series_data	array[array]	2D array of floats representing the chronological time series points.

EXAMPLE REQUEST BODY

```
{
  "ar_number": 12673,
  "model_type": "lstm_mvts_v2",
  "features": ["TOTUSJH", "TOTBSQ"],
  "time_series_data": [
    [1542.12, 102345.1],
    [1544.89, 102489.3]
  ]
}
```

EXAMPLE RESPONSE

200 OK

```
{
  "ar_number": 12673,
  "prediction": "X-Class",
  "probabilities": {
    "B-Class": 0.01,
    "C-Class": 0.04,
    "M-Class": 0.12,
    "X-Class": 0.83
  },
  "latency_ms": 142
}
```

To facilitate public and remote access to our automated solar flare analysis and forecasting framework, the backend REST API is hosted on the server infrastructure at the National Institute of Science Education and Research (NISER), Bhubaneswar. Accessible via the endpoint `/api/v1/predict`, the API allows researchers to submit multivariate time series (MVTs) data from SHARP parameters for a specific Active Region (AR) using a POST request containing parameters such as `ar_number`, `model_type`, `features`, and `time_series_data` to receive flare classification probabilities and predictive analysis.

REST API – Everyone

Pipeline

1. **Physics-Based Coupling (1D):** Extracts explicit features (Transfer Entropy, Dynamic Time Warping, Cross-Correlation, Hardness Ratio) between HXR and SXR light curves.
2. **Time-Frequency Analysis (2D):** Transforms X-ray signals using Wavelet analysis to capture non-stationary precursor signatures and quasi-periodic pulsations.
3. **Multimodal Transformer:** Fuses the 1D coupling indicators with 2D wavelet maps utilizing a novel cross-attention mechanism.
4. **Forecasting Engine:** Outputs the precise probability, class, and lead time of impending solar flares.
5. **Model Training:** Once we retrieved the clean data we trained it using Time Series Models and got the accuracy of 95.31%.

Novelty

Current techniques for solar flare classification are manual and totally relied on physics.

For automation we trained last 10 years data from NASA SDAC and 2 layer Xray data from Astrostat Helio data.

This helps us to automate the classification for the solar physicist (Python Package) and the climate scientist (Web Portal) for fast and detailed classification.



7. TECH Stack / Budget

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Tech Stack

1. Python
2. PYPI
3. REST API
4. MERN
5. FASTAPI
6. Firebase
7. Hugging Face
8. Docker
9. Twin

[Deep Learning, RESNET, Time Series Models, TORCH, SUNPY, ASTROPY, MATPLOTLIB, NUMPY, ETC]

Estimated Budget

We have used the HPC of NISER BHU for training our data because Jashanpreet (Team Member) have the access to it.

Hence the estimated cost of building the model is **NIL**. However there will be a minor annual costing for the hosting the website.



8. Deliverable / Expected Outcome

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NEW UPDATE: A new robust LSTM model for M and X class flares is available! Note: It is not in the stable package yet, but you can [download weights here](#).



-- an open source python package for solar flare analysis

SunFLAAR is an open-source Python package for **comprehensive solar flare analysis**, integrating deep learning and statistical methods. It provides automated tools for **multivariate time-series forecasting**, **active region classification**, and **imbalanced data metrics** to aid space weather research.

Please remember to **acknowledge and cite** the use of SunFLAAR in your publications.

Interface Options

Web Portal

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About

Learn about SunFLAAR, its background architectures and modules.



API Reference

REST API endpoints, command reference, and prediction usage.

0 | 0

LATEST RELEASE

1.0.4

[Check Release Notes](#)

<https://sunflaar.vercel.app>

<https://sunflaar.vercel.app>

<https://pypi.org/project/sunflaar/>

astrodingra@gmail.com

9. Team Role

Role → Team ↓	Deep Learning	Physics Research	Web Portals	PYPI Package	IUCAA/NISER PEGASUS CLUSTER
Tegkirat Kaur			✓	✓	
Jashanpreet Singh	✓	✓		✓	✓

Work was distributed in 5 major categories including the model training and model implication which include the background research fo solar physics and publishing the package at PYPI to make that available as the python package for researchers.

Purpose	Reference
SHARP + LSTM temporal forecasting	Wang et al. 2020
SHARP + GOES + LSTM intensity	Jiao et al. 2020
GOES time-series CNN	Landa & Reuveni 2021
B/C/M/X classification	Abduallah et al. 2021
Operational/deep-learning forecasting	Chen et al./related CNN work
Particle release-time methodology	Agueda et al. 2014
VDA/release-time limitations	Wang & Qin 2013
Modern SEP release analysis	GLE74 Solar Physics paper
Current deep-learning state of the art	Riggi et al. 2026



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WEB PORTAL

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thank you!

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